S.B. JAIN INSTITUTE OF TECHNOLOGY MANAGEMENT & RESEARCH, NAGPUR

Practical 09

Aim: A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Optimal page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

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❖ CODE :

```
~
 GNU nano 8.7
#include <stdio.h>

int main() {
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int frame[3], n=12, frames=3;
    int i,j,k,hit=0,fault=0,pos,far;

    for(i=0;i<frames;i++) frame[i]=-1;

    for(i=0;i<n;i++) {
        int found=0;

        for(j=0;j<frames;j++)
            if(frame[j]==pages[i]) { hit++; found=1; break; }

        if(!found) {
            fault++;

            for(j=0;j<frames;j++) {
                if(frame[j]==-1) { frame[j]=pages[i]; found=1; break; }
            }

            if(!found) {
                far=-1;
                for(j=0;j<frames;j++) {
                    int next=-1;
                    for(k=i+1;k<n;k++)
                        if(frame[j]==pages[k]) { next=k; break; }

                    if(next==-1) { pos=j; break; }

                    if(next>far) { far=next; pos=j; }
                }
                frame[pos]=pages[i];
            }
        }
    }

    printf("Hits=%d\nFaults=%d\n",hit,fault);
    printf("Hit%%=%.2f\nFault%%=%.2f\n",(float)hit/n*100,(float)fault/n*100);

    return 0;
}

^G Help        ^O Write Out    ^F Where Is     ^K Cut          ^T Execute
^X Exit        ^R Read File    ^\ Replace      ^U Paste        ^J Justify
```

❖ OUTPUT :

```
~
Dell@DESKTOP-D7MT61A UCRT64 ~
$ nano optimal.c

Dell@DESKTOP-D7MT61A UCRT64 ~
$ gcc optimal.c -o optimal

Dell@DESKTOP-D7MT61A UCRT64 ~
$ ./optimal
Hits=5
Faults=7
Hit%=41.67
Fault%=58.33

Dell@DESKTOP-D7MT61A UCRT64 ~
$
```